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$$\int e^{\sqrt{x}} dx$$

$$\text{stel } t = \sqrt{x}$$

$$t^2 = x$$

$$2t dt = dx$$

$$= 2 \int t e^t dt$$

$$\begin{cases} u = t \\ dv = e^t dt \end{cases} \Rightarrow \begin{cases} du = dt \\ v = e^t \end{cases}$$

$$= 2te^t - 2 \int e^t dt$$

$$= 2te^t - 2e^t + C$$

$$= 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$$

References